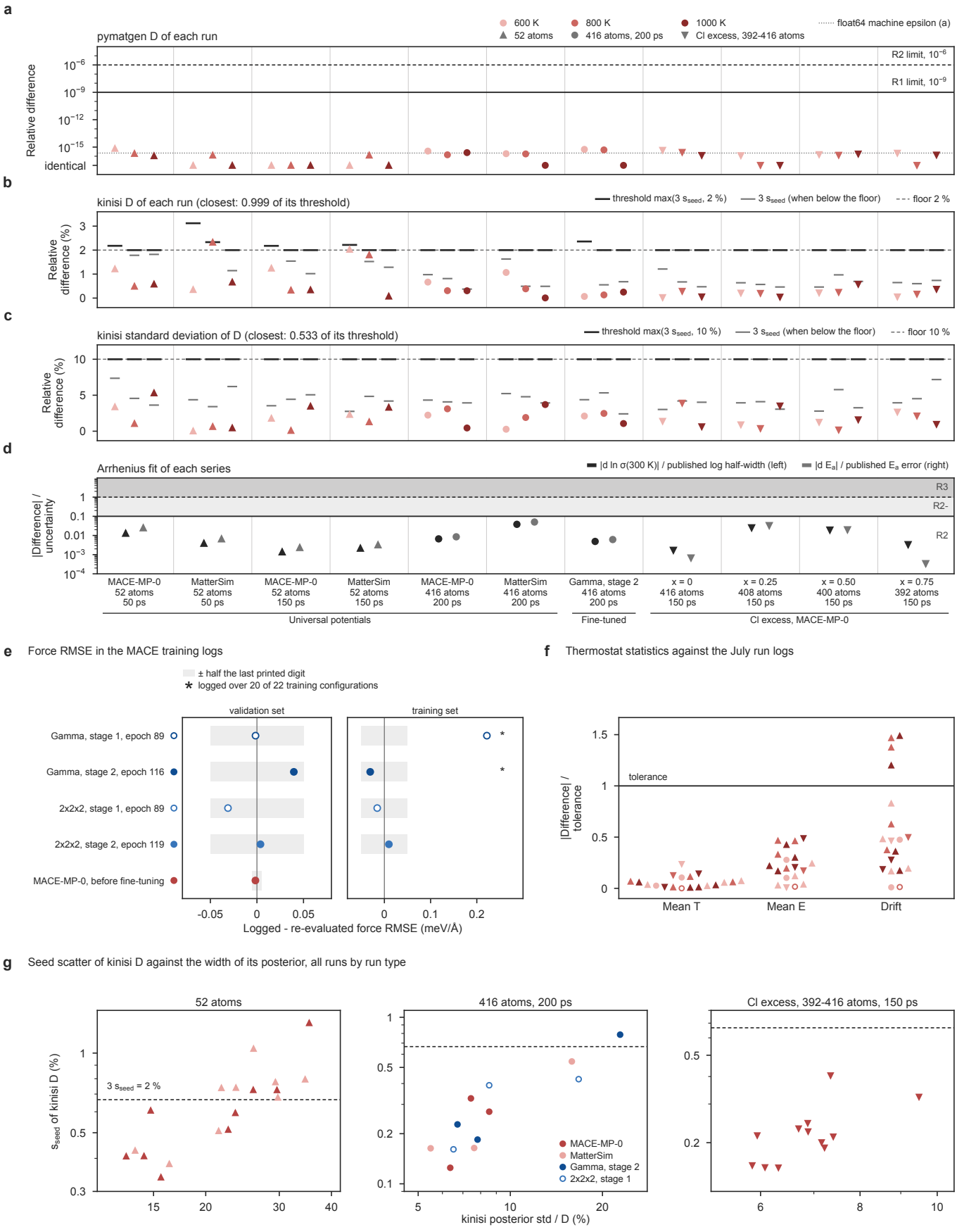

a
pymatgen D of each run
600 K
800 K
1000 K
float64 machine epsilon (a)
52 atoms
416 atoms, 200 ps
CI excess, 392-416 atoms
Relative difference
R2 limit, 10⁻⁶
R1 limit, 10⁻⁹
identical
b
kinisi D of each run (closest: 0.999 of its threshold)
threshold max(3 s_seed, 2 %)
3 s_seed (when below the floor)
floor 2 %
Relative difference (%)
c
kinisi standard deviation of D (closest: 0.533 of its threshold)
threshold max(3 s_seed, 10 %)
floor 10 %
d
Arrhenius fit of each series
|d ln σ(300 K)| / published log half-width (left)
|d E_a| / published E_a error (right)
|Difference| / uncertainty
R3
R2-
R2
MACE-MP-0 52 atoms 50 ps
MatterSim 52 atoms 50 ps
MACE-MP-0 52 atoms 150 ps
MatterSim 52 atoms 150 ps
MACE-MP-0 416 atoms 200 ps
MatterSim 416 atoms 200 ps
Gamma, stage 2 416 atoms 200 ps
x = 0 416 atoms 150 ps
x = 0.25 408 atoms 150 ps
x = 0.50 400 atoms 150 ps
x = 0.75 392 atoms 150 ps
Universal potentials
Fine-tuned
CI excess, MACE-MP-0
e
Force RMSE in the MACE training logs
± half the last printed digit
* logged over 20 of 22 training configurations
validation set
training set
Gamma, stage 1, epoch 89
Gamma, stage 2, epoch 116
2x2x2, stage 1, epoch 89
2x2x2, stage 2, epoch 119
MACE-MP-0, before fine-tuning
Logged - re-evaluated force RMSE (meV/Å)
f
Thermostat statistics against the July run logs
|Difference| / tolerance
tolerance
Mean T
Mean E
Drift
g
Seed scatter of kinisi D against the width of its posterior, all runs by run type
52 atoms
416 atoms, 200 ps
CI excess, 392-416 atoms, 150 ps
s_seed of kinisi D (%)
3 s_seed = 2 %
MACE-MP-0
MatterSim
Gamma, stage 2
2x2x2, stage 1
kinisi posterior std / D (%)